

A Novel Heuristic Data Routing for Urban Vehicular *Ad Hoc* Networks

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Abstract—This work is devoted to solving the problem of multi-criteria multihop routing in vehicular *ad hoc* networks (VANETs), aiming at three goals: 1) increasing the end-to-end delivery ratio; 2) reducing the end-to-end latency; and 3) minimizing the network overhead. To this end and beyond the state of the art, heuristic routing for vehicular networks (HERO), which is a distributed routing protocol for urban environments, encapsulating two main components, is proposed. The first component, road-segment selection, aims to prioritize the road segments based on a heuristic function that contains two probability distributions, namely, shortest distance distribution (SDD) and connectivity distribution (CD). The mass function of SDD is the product of three quantities: 1) the perpendicular distance; 2) the dot-product angle; and 3) the segment length. On the other hand, the mass function of CD considers two quantities: 1) the density of vehicles and 2) the interdistance of vehicles on the road segment. The second component, vehicle selection, aims to prioritize the vehicles on the road segment based on four quantities: 1) the relative speed; 2) the movement direction; 3) the available buffer size; and 4) signal fading. The simulation results showed that HERO achieved a promising performance in terms of delivery success ratio, delivery delay, and communication overhead.

Index Terms—Heuristic routing, intelligent transportation systems (ITSs), probabilistic routing, vehicular *ad hoc* networks (VANETs).

I. INTRODUCTION

VEHICULAR communication can play a critical role in permitting numerous applications, such as traffic control, collision avoidance, and lane-change assistance in intelligent transportation systems (ITSs) [1]. This communication paradigm paves the way for many applications to realize

the future of ultimately autonomous driving. Indeed, efficient vehicular networking is fundamental to a wide range of applications in road safety, business, infotainment, and smart cities [2], [3]. Generally, vehicular *ad hoc* networks (VANETs) consist of a limited number of vehicles, each equipped with an onboard unit (OBU) together with GPS and street-level *digital maps*. OBU enables intervehicle communication, vehicle-to-vehicle (V2V), as well as communication between roadside units (RSUs) and vehicles, sometimes named vehicle to infrastructure (V2I). In particular, fewer RSUs are deployed to facilitate cost-effective vehicular communication [4]. Nevertheless, RSUs are still not yet widely available and have experienced a very slow pace of implementation due to their expensive cost [5]. RSU installation and placement strategies are studied in [6]–[8].

The mainstay for implementing ITS applications is the multihop routing mechanism, which should guarantee the quality of services such as higher packet delivery ratio, lower latency, and smaller overhead. Multihop routing is an appropriate mechanism to disseminate data in such mobile-node networks [9]–[11]. In such a mechanism, the determination of a set of road segments and a set of relay vehicles is the main research issue [12]. The process of selecting a set of road segments is called *interrouting*, while the process of selecting a set of relay vehicles is called *intrarouting*. Both *interrouting* and *intrarouting* employ different routing criteria that mostly characterize the quality of the selected path between source and destination vehicles. Shortest distance and density are the frequent criteria that have been used to estimate the quality of *interrouting*, while the speed and direction are usually employed to characterize the *intrarouting*. Although the multihop routing for mobile environments has been extensively investigated in the context of the mobile *ad hoc* network (MANET) [13], the vehicular network has obvious features, making the routing mechanism in MANET no longer suitable. VANET possesses unique characteristics such as energy is no longer an issue, and nodes have higher mobility in restricted streets.

Problem and Motivations: The network-layer performance in VANETs is conjointly influenced by multiple quantities (i.e., criteria), such as vehicle's speed, signal fading, direction angle between the source and next-hop vehicles, the shortest distance from the source to destination, communication range, road size (i.e., number of lanes), segment length, segment connectivity, etc. Previous works considered few criteria during the selection of forwarder vehicles or road segments. For example, Lin *et al.* [14] prioritized the next-hop forwarders based on

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vehicle's movement similarity by considering two quantities: 1) the speed and 2) the movement direction. In contrast, Yao *et al.* [15] prioritized the next-hop forwarders based on the shortest distance to the next-hop junction. More quantities and examples are explained in Section II. Such limited quantities do not thoroughly actualize the main parameters that influence network's layer in VANETs. The performance of VANET is mostly evaluated through three essential metrics, namely, end-to-end latency, delivery ratio, and communication overhead. These metrics are interrelated with multiple criteria. Latency is proportionally correlated to the distance between the source and destination. Longer distance leads in general to more hops, more junctions from the source to the destination, and more carry-and-forward times. This definitely results in more network's partitions and upsurges the latency. Despite it is desirable for the data packet to travel through shorter distance segments, the shortest distance segments are not always an efficient selection because the connectivity of such segments is not constantly guaranteed. Higher density inevitably improves network connectivity and reduces the number of carry-and-forwards, thereby reducing latency. Moreover, higher density allows packets to travel primarily on selected segments, thus reducing switching failures of packets at junctions and reducing the latency as well. Likewise, a longer distance decreases the delivery ratio because packets may encounter network's partitions, which is considered as the main reason for packets dropping. Longer distance also has an impact on communication overhead because it is necessary to select more forwarders through multihops, and then more operations need to be coordinated all the way. Furthermore, the process of packet switching is increased with longer distance, which in turn increases the overhead, especially when the packet fails to be switched. Vehicle's speed, movement direction, and other quantities such as signals fading and communication range had an impact on network performance. These quantities entirely captured the performance of VANETs, and are deeply explained and mathematically modeled in Section IV. This motivated us to design heuristic routing for vehicular networks (HERO), which considers seven quantities to select the next-hop segment (interrouting) and four quantities to select the next-hop vehicle (intrarouting).

Contributions: We developed a routing protocol, namely, HERO, which operates in a distributed manner to heuristically select the road segments and the relay vehicles. The selection of road segments is based on two heuristic functions. The first heuristic function aims to select the *Shortest Distance* by aggregating three quantities: 1) dot production angle; 2) perpendicular distance; and 3) segment length. The second heuristic function considers the segment connectivity through four quantities: 1) the range of communication; 2) lanes count; 3) segment length; and 4) the mean vehicle count. On the other hand, the selection of relay vehicles is attained by considering four quantities: 1) the moving direction of the vehicle; 2) speed difference; 3) signal fading; and 4) buffer size. Simulation and documentation of HERO are available online in the link.¹

Organization: The remainder of this article is organized as follows. Section II explains previous studies. Section III explains the preliminaries, including the traffic and propagation models. Section IV explains the proposed protocol, while the performance analysis is elaborated in Section V. The experiments and discussions are explained in Section VI. Finally, Section VII concludes this work.

II. RELATED WORK

From the perspective of the network structure, the routing protocols are classified into clustered-structure and flat-structure. Moving zone (MOZO) [14] is an example of clustered based, which partitioned the road network into disjoint zones according to the similarity of movements, which is actually computed by considering multiple criteria, such as speed, timestamp, and the direction of movement. Each zone has a header vehicle that manages the member vehicles in the zone as well as dissimilates the data packets. BRAVE is an example of the flat structure, introduced in [16], it employed an opportunistic scheme, in which the relay vehicles are selected based on the shortest distance. BRAVE operates in two main steps: 1) opportunistic data forwarding and 2) spatial awareness. In spatial awareness, BRAVE selects the road segment by employing Dijkstra's algorithm, in which network information and digital map street are utilized. In the second step, the packets are opportunistically routed on the selected road segment.

Furthermore, from the perspective of network infrastructures, the routing protocols in VANET are classified into pure V2V routing and V2V assisted by RSUs routing (V2I). Due to the slow deployment of RSUs, most of the studies consider the V2V routing strategy, e.g., [12], [14], and [16]–[18]. However, recently, a few studies have considered V2I, e.g., [5], [15], and [19]–[21]. An example of V2V is the density-aware emergency message extension protocol (DEEP) [18]. DEEP divides roads into multiple equal-sized blocks based on vehicle density and assigns different priorities to vehicles. Each block is assumed to contain one vehicle. The farther block has a shorter delay time than the closer block to forward the emergency messages. Low density results in larger block sizes, and as the density becomes higher, the block size becomes smaller. On the other hand, an example of V2I is the *hidden Markov model* (PRHMM) [15]. PRHMM supports V2V and V2I; it predicts the upcoming vehicle's location based on its historical mobility patterns and predicts the movement pattern toward the destination using the *hidden Markov model*. The vehicle that has a higher probability of reaching the destination is selected as a next hop in the path.

Map-based protocols utilize GPS and digital maps to select the routing path between the source and the destination. Examples of *map-based* routing are addressed in [12] and [17]. In [17], a street-centric routing protocol, called the street-centric routing protocol based on microtopology (SRPMT), was proposed. SRPMT is designed based on the concept of *microtopology* (i.e., the road segment between two consecutive intersections), which considers the mobility of the vehicle, the signal fading, the wireless channel contention, and existing data traffic. SRPMT runs in two steps:

¹<https://github.com/howbani/VSIM>

TABLE I
NOTATIONS

Notation	Description
n_i	Vehicle i , located at (x_i, y_i) .
$d_{i,j}$	The distance between the vehicles n_i and n_j .
δ	Denotes the communication range of OBU.
$v_i, (x_i, y_i)$	$\mathcal{V}_G = \{v_0, v_1, v_2, \dots, v_i\}$ represents road junctions. v_s denotes the source junction. The destination junction is denoted by v_d . (x_i, y_i) denotes the location of junction v_i .
$r_{ij}, r_{ij} $	The road segment that joins two junctions $v_i, v_j \in \mathcal{V}_G$. Its length is denoted by $ r_{ij} $.
$E_{ij}^R, E_{ij}^S, E_{ij}^T$	E_{ij}^R denotes the expected number of vehicles residing within the segment r_{ij} . E_{ij}^S denotes the average of vehicle speed allowed in the segment r_{ij} . E_{ij}^T denotes the average time that the vehicle resides in r_{ij} .
s_{ij}^{max}	The maximum speed allowed in the segment r_{ij} .
α_{ij}	Interarrival time of vehicles for the segment r_{ij} .
$\theta_{i,j}$	The direction angle between the junctions v_i and v_j .
$\Phi_{i,j}$	The distance distribution between the junctions v_i and v_j .
λ_{ij}	$\lambda_{ij} = 1/\alpha_{ij}$ is the arrival rate of vehicles to the segment r_{ij} .
$\tilde{\phi}_{ij}, \tilde{Q}_{i,j}$	The heuristic function of <i>Inter-Path</i> and the heuristic function of <i>Intra-Path</i> , respectively.
$\tilde{\psi}_{i,j}, \tilde{L}_{i,j}$	The perpendicular distance distribution and road segment length distribution, respectively.
$\tilde{S}_{i,j}, \tilde{D}_{i,j}$	Speed difference distribution, moving direction
$\tilde{M}_{i,j}, \tilde{F}_{i,j}$	distribution, buffer size distribution and signal fading distribution, respectively.
$\tilde{X}_{i,j}$	Mobility similarity score.

1) the selection of the next successive road segments based on *Dijkstra's algorithm* and 2) the packet routing within the road segment based on an opportunistic metric that considers the end-to-end delay. An opportunistic metric [expected transmission cost over a multihop path (ETCoP)] is proposed in [12]. ETCoP is based on *link correlation*, which mathematically expresses how the transmission cost of a given packet is influenced when distinct links are utilized. Based on ETCoP, the authors designed SRPE (street-centric opportunistic routing protocol), which runs in two steps as in SRPMT.

Adaptive multiple-hop routing protocols are proposed in [22] and [23]. Both are designed based on *fuzzy logic*, which is an appropriate computational model, especially for networks with rapid topology variations. PFQ-AODV [22] utilizes fuzzy constraint Q-learning to acquire the routing paths based on AODV routing. Fuzzy logic is employed to evaluate the quality of wireless link through considering multiple criteria such as bandwidth and vehicle movement. Hawbani *et al.* [23] modeled the routing problem as *MCDM*, in which the attributes, such as the transmission distance, density, and other attributes that have an influence on the network layer are expressed by fuzzy sets, which are embodied by *fuzzy membership functions*. The final decision is derived by using the TSK inference system.

Unlike the previous works, this article captures multiple quantities when packets are routed. Our work selects road segments with a shorter distance and higher connectivity and prioritizes the vehicles in the selected road segment based on

the speed difference, the vehicle's moving direction, buffer size, and signal fading.

III. PRELIMINARIES

This section explains the propagation and traffic models used in this article. Notations, which are frequently used in this article, are listed in Table I.

A. Propagation Model

We employed *Nakagami- m propagation model* as formulated in (1), where $m \geq 1/2$ denotes the path loss while Ω represents the *average power strength*, [24], [25]. The CDF of *Nakagami- m* model is formulated as in (2). The packet is received successfully when the probability of the received power is larger than the value r_x as expressed in (3). Both parameters Ω and r_x are obtained from the *free-space model* as expressed in (4), where T_p denotes the transmission power, and $d_{i,j}$ denotes the *Euclidian* distance from a sender vehicle n_i to a receiver vehicle n_j . The notations δ , G_t , G_r and λ denote the range of communication, the gain of antenna for the transmitter, the antenna gain for the receiver, and the wavelength of the signal, respectively. Finally, the value of m is derived by (5) as in [26]

$$f_m(x; m, \Omega) = \frac{m^m x^{m-1} e^{-(m/\Omega)x}}{\Omega^m (m-1)!} \quad (1)$$

$$F_m(x; m, \Omega) = \int_0^x f_m(z; m, \Omega) dz$$

$$= \frac{m^m}{\Omega^m (m-1)!} \int_0^x z^{m-1} e^{-(m/\Omega)z} dz \quad (2)$$

$$Pr(x > r_x) = 1 - F_m(r_x; m, \Omega)$$

$$= e^{-\frac{m}{\Omega} r_x} \sum_{i=0}^m \frac{(\frac{m}{\Omega} r_x)^{i-1}}{(i-1)!} \quad (3)$$

$$\Omega(d_{i,j}) = \frac{T_p}{d_{i,j}^2} G; \quad r_x(\delta) = \frac{T_p}{\delta^2} G = \frac{G_t G_r \lambda^2}{16\pi^2} \quad (4)$$

$$m = \begin{cases} 1.0, & d_{i,j} \geq 150\text{m} \\ 1.5, & 50\text{m} \leq d_{i,j} < 150\text{m} \\ 3.0, & d_{i,j} < 50\text{m}. \end{cases} \quad (5)$$

B. Road Network and Traffic Model

The road network that contains $|\mathcal{V}_G|$ junctions or intersections and $|\mathcal{E}_G|$ road segments can be modeled by a directed graph $\mathcal{G} = (\mathcal{V}_G, \mathcal{E}_G)$, such that the junctions represent the vertex set $\mathcal{V}_G = \{v_0, v_1, v_2, \dots, v_i\}$, while the road segments (multilanes as shown in Fig. 1) represent the edge set $\mathcal{E} \subseteq \mathcal{V}_G \times \mathcal{V}_G$. Let $r_{i,j}$ be the road segment that joins the junctions $v_i, v_j \in \mathcal{V}_G$. For a given road segment $r_{i,j}$, the *interarrival time* (in seconds) of two successive arrivals is modeled by the *exponential distribution* with the parameter $\alpha_{i,j} > 0$. This means that any arrival occurring at any given time is independent of the length of time that has elapsed from the previous arrival [5], [27]. The PDF of the exponential distribution is

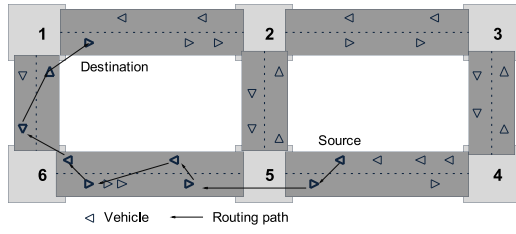


Fig. 1. Road network.

formulated by (6). For a given road network $\mathcal{G} = (\mathcal{V}_\mathcal{G}, \mathcal{E}_\mathcal{G})$ with $|\mathcal{E}_\mathcal{G}|$ segments, each segment $r_{i,j}$ has different *interarrival mean* $\alpha_{i,j}$ due to some factors such as the time or the location of the segment in the city. At the simulation level, the mean value of $\alpha_{i,j}$ is obtained by the *Box-Muller Transform* (7), where μ_I and σ_I denote the *mean* and *deviation* of the *Gaussian* distribution, respectively

$$I_{i,j}(t, \alpha_{i,j}) = \alpha_{i,j} e^{-t\alpha_{i,j}} \quad (6)$$

$$\alpha_{i,j} = \left| \mu_I + \left(\sigma_I \sqrt{-2\log z_0} \sin(2\pi z_1) \right) \right| \quad (7)$$

$$z_0, z_1 \in [0, 1] \quad \forall r_{i,j} \in \mathcal{E}_\mathcal{G}.$$

For a given road segment, the arrival of vehicles follows the Poisson process with a rate $\lambda_{i,j} = 1/\alpha_{i,j}$. Thus, the probability that k vehicles arrive at the entry of $r_{i,j}$ in one second is obtained by the *probability mass function*

$$A_{i,j}(k) = \frac{\lambda_{i,j}^k}{k!} e^{-\lambda_{i,j}}. \quad (8)$$

The probability that k vehicles reside within $r_{i,j}$ is given by the mass function (9), where $|r_{i,j}|$ denotes the segment length, and $\rho_{i,j} = \lambda_{i,j}|r_{i,j}|$ is the density of road segment $r_{i,j}$ (vehicles per segment) [5]. $E_{i,j}^S$ is the average speed of vehicles allowed in segment $r_{i,j}$, formulated by (10) where $\bar{t}$ is the average length of a vehicle, and $s_{i,j}^{\max}$ is the maximum speed allowed in segment $r_{i,j}$. $E_{i,j}^R$ denotes the expected number of vehicles residing within segment $r_{i,j}$, which is obtained from Little's law (11), where $E_{i,j}^T = |r_{i,j}|/E_{i,j}^S$ is the average time that the vehicle resides in $r_{i,j}$

$$R_{i,j}(k) = \frac{(\rho_{i,j})^k}{k!} e^{-\rho_{i,j}} \quad (9)$$

$$E_{i,j}^S = s_{i,j}^{\max} \left(1 - \frac{\bar{t} E_{i,j}^R}{|r_{i,j}|} \right) \quad (10)$$

$$E_{i,j}^R = \lambda_{i,j} E_{i,j}^T \approx \sum_{x=0}^{|r_{i,j}|/\bar{t}} x R_{i,j}(x). \quad (11)$$

At the simulation level, for a given segment $r_{i,j} \in \mathcal{E}_\mathcal{G}$, the maximum speed $s_{i,j}^{\max}$ is obtained by the *Box-Muller Transform* (12), where μ_S and σ_S are the *mean* and the *deviation* of the *Gaussian* distribution, respectively

$$s_{i,j}^{\max} = \left| \mu_S + \left(\sigma_S \sqrt{-2\log z_0} \sin(2\pi z_1) \right) \right| \quad (12)$$

$$z_0, z_1 \in [0, 1] \quad \forall r_{i,j} \in \mathcal{E}_\mathcal{G}.$$

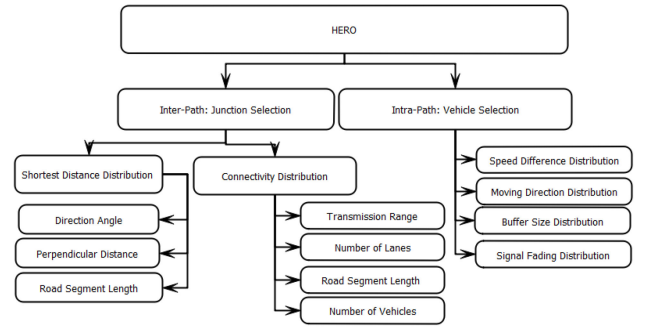


Fig. 2. Main components of HERO.

The per-vehicle speed is modeled by a random variable that follows *truncated normal distribution* with density function

$$S_{i,j}(s_k) = \frac{1}{\sigma_S \sqrt{2\pi}} e^{\left(\frac{s_k - E_{i,j}^S}{\sigma_S \sqrt{2}} \right)^2} \bigg/ \int_{s_{i,j}^{\min}}^{s_{i,j}^{\max}} \frac{1}{\sigma_S \sqrt{2\pi}} e^{\left(\frac{s_k - E_{i,j}^S}{\sigma_S \sqrt{2}} \right)^2}. \quad (13)$$

IV. PROPOSED PROTOCOL

For a given junction $v_i \in \mathcal{V}_\mathcal{G}$, let $\mathcal{V}_i = \{v_j | v_j \in \mathcal{V}_\mathcal{G} \& r_{i,j} \in \mathcal{E}_\mathcal{G}\}$ be the set of adjacent junctions of v_i . For example, in Fig. 1, $\mathcal{V}_2 = \{v_1, v_3, v_5\}$. The position of v_i is denoted by (x_i, y_i) . The source junction is denoted by v_s while the destination junction is denoted by v_b .

To achieve the three goals of this work, we propose a solution that subdivided the problem into two main components; each is characterized by a heuristic function as depicted in Fig. 2. The two heuristic functions: 1) *Interpath* and 2) *Intrapath* are proposed to improve the selection of road segments and relay vehicles, respectively. *Interpath* is designed to select a routing path that consists of multiple road segments with a shorter distance and higher connectivity. This is implemented through two probability distributions: 1) shortest distance distribution (SDD) and 2) the connectivity distribution (CD). SDD considers three quantities: 1) direction angle; 2) perpendicular distance; and 3) segment length, while CD considers four quantities: 1) the communication range; 2) lane count; 3) the length of segment; and 4) the expected vehicle count. On the other hand, *Intrapath* prioritizes the relay vehicles based on four quantities: 1) speed difference; 2) vehicle's moving direction; 3) buffer size; and 4) signal fading. All these quantities are explained thoroughly in the following sections. An illustrative numerical example for *IntraPath* and *InterPath* are involved in the supplementary file.²

A. InterPath: Junction Selection

An *Interpath* is a sequence of consecutive junctions connecting the source to the destination junctions, heuristically satisfying two key requirements: 1) shorter routing-distance and 2) higher connectivity. The selection decision is aggregated through weighted averaging $\tilde{\mathcal{P}}_{i,j}$ (14) with two probability

²<https://github.com/howbani/VSIM/blob/master/suppl.pdf>

Algorithm 1 Road Segment Selection**Input:** Current junction v_i ; destination junction v_b .**Output:** The road segment r_{ij} with the highest priority.

```

1. JunctionSelection ( $v_i, v_b$ )
2. {
3.    $\forall v_j \in \mathcal{V}_i$  Do
4.   {
5.     Compute  $\tilde{\theta}_{i,j}$  by Eq. (17). // Dot angle production.
6.     Compute  $\tilde{\phi}_{i,j}$  by Eq. (20). // Perpendicular
       distance.
7.     Compute  $\tilde{\mathcal{L}}_{i,j}$  by Eq. (22). // Segment length.
8.     Compute  $\tilde{\psi}_{i,j}$  by Eq. (15). // Shortest Distance
       Distribution.
9.     Compute  $\tilde{\xi}_{i,j}$  by Eq. (26). // Connectivity
       Distribution.
10.   }
11.    $\forall v_j \in \mathcal{V}_i$  Do
12.     Compute  $\tilde{P}_{ij}$  by Eq. (14). // Segment priority.
13.   return  $r_{ij}$  with the max  $\tilde{P}_{ij}$ .
14. }
```

distributions: 1) CD [denoted by $\tilde{\xi}_{i,j}$ (26)] and 2) the SDD [denoted by $\tilde{\Phi}_{i,j}$ (15)]. The priority of the selection (i.e., for the junction $v_i \in \mathcal{V}_g$) is computed in a distributed manner as expressed in (14) where w_ϕ and w_ξ are tuning parameters

$$\tilde{P}_{i,j} = w_\phi \tilde{\Phi}_{i,j} + w_\xi \tilde{\xi}_{i,j} \Big/ \sum_{\forall v_x \in \mathcal{V}_i} w_\phi \cdot \tilde{\Phi}_{i,x} + w_\xi \tilde{\xi}_{i,x} \quad \forall v_j \in \mathcal{V}_i$$

$$w_\phi + w_\xi = 1. \quad (14)$$

Algorithm 1 outlines the selection of the next segment, given that a vehicle is heading toward a junction v_i .

B. Shortest Distance Distribution

The SDD $\tilde{\Phi}_{i,j}$ is obtained by combining three *probability mass functions*, formalized in (15), including the *direction angle* $\tilde{\theta}_{i,j}$, the *perpendicular distance* $\tilde{\psi}_{i,j}$ and the segment's length $\tilde{\mathcal{L}}_{i,j}$. The direction distribution assigns higher priority for junctions that are toward the destination junction, while the perpendicular distance distribution assigns higher priority for junctions that are closer to the centerline between the source and the destination (see Fig. 3). The shortest distance is computed by summing the perpendicular distance and segment length, all multiplied by the direction. This ensures that as long as the direction is correct, both the perpendicular distance and segment length will have an impact. Obviously, the direction has the greatest impact, while both perpendicular distance and

segment length have an equal impact

$$\tilde{\Phi}_{i,j} = \tilde{\theta}_{i,j} (\tilde{\psi}_{i,j} + \tilde{\mathcal{L}}_{i,j}) \Big/ \sum_{\forall v_x \in \mathcal{V}_i} \tilde{\theta}_{i,x} (\tilde{\psi}_{i,x} + \tilde{\mathcal{L}}_{i,x}) \quad \forall v_j \in \mathcal{V}_i. \quad (15)$$

Direction Angle: The direction angle $\theta_{i,j}$ between the current junction v_i and the potential next-junction v_j toward the destination junction v_b is modeled as a dot product of two vectors $\vec{a} \cdot \vec{c}$, such that $\vec{a} = (x_j - x_i, y_j - y_i)$ and $\vec{c} = (x_b - x_i, y_b - y_i)$, and normalized to π by (16), shown at this bottom of the page. The normalized-direction $\bar{\theta}_i = (\bar{\theta}_{i,1}, \bar{\theta}_{i,2}, \dots, \bar{\theta}_{i,b_i})$ is injected into the mass function (17), shown at the bottom of the page, to obtain the random variable $\tilde{\theta}_i = (\tilde{\theta}_{i,1}, \tilde{\theta}_{i,2}, \dots, \tilde{\theta}_{i,b_i})$. Direction distribution assigns a higher probability for the normalized angles $0 \leq \bar{\theta}_{i,j} \leq (1/2)$ that ensure higher routing progress toward the destination junction.

Perpendicular Distance: The perpendicular distance distribution $\tilde{\psi}_{i,j}$ allocates a higher probability for the junctions that are closer to the central line $l_{s,b}$ [i.e., a virtual line linking the source junction v_s and the destination junction v_b (see Fig. 3) Assume that v_4 is the current junction, $\mathcal{V}_4 = \{v_s, v_2, v_3, v_5, v_6, v_7\}$. The two-head arrow indicates the central line $l_{s,b}$ and dashed lines indicate the *perpendicular distance* for junctions in $\mathcal{V}_4$. The perpendicular distance ψ_j from v_j to $l_{s,b}$ is defined by (18). For source junction v_i , we define the normalized perpendicular-distance random variable $\bar{\psi}_i = (\bar{\psi}_{i,1}, \bar{\psi}_{i,2}, \dots, \bar{\psi}_{i,b_i})$ by (19). Furthermore, we define the perpendicular-distance probability distribution, denoted by $\tilde{\psi}_i = (\tilde{\psi}_{i,1}, \tilde{\psi}_{i,2}, \dots, \tilde{\psi}_{i,b_i})$, by (20)

$$\psi_j = \frac{|x_j(y_b - y_s) - y_j(x_b - x_s) + x_b y_s - y_b x_s|}{\sqrt{(x_b - x_s)^2 + (y_b - y_s)^2}} \quad (18)$$

$$\bar{\psi}_{i,j} = \frac{\psi_j}{\sum_{\forall v_x \in \mathcal{V}_i} \psi_x} \quad \forall v_j \in \mathcal{V}_i \quad (19)$$

$$\tilde{\psi}_{i,j} = e^{-\bar{\psi}_{i,j}} / \sum_{\forall v_x \in \mathcal{V}_i} e^{-\bar{\psi}_{i,x}} \quad \forall v_j \in \mathcal{V}_i. \quad (20)$$

Road Segment Length: This distribution allocates a higher probability to the longer segment. This is because the longer segments are preferable to avoid frequent segment switching that leads to hinder packet delivery, especially when the vehicle density increases. For the source junction v_i , we define the normalized segment length random variable $\bar{\mathcal{L}}_i = (\bar{\mathcal{L}}_{i,1}, \bar{\mathcal{L}}_{i,2}, \dots, \bar{\mathcal{L}}_{i,b_i})$ by (21). Furthermore, we define

$$\bar{\theta}_{i,j} = \cos^{-1} \left(\frac{(x_j - x_i)(x_b - x_i) + (y_j - y_i)(y_b - y_i)}{\sqrt{(x_b - x_i)^2 + (y_b - y_i)^2} \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}} \right) \Big/ \pi \quad (16)$$

$$\tilde{\theta}_{i,j} = \begin{cases} \left(1 - \frac{\bar{\theta}_{i,j} e^{\bar{\theta}_{i,j}}}{1 + \bar{\theta}_{i,j} e^{\bar{\theta}_{i,j}}}\right)^\tau \Big/ \sum_{\forall v_x \in \mathcal{V}_i} \left(1 - \frac{\bar{\theta}_{i,x} e^{\bar{\theta}_{i,x}}}{1 + \bar{\theta}_{i,x} e^{\bar{\theta}_{i,x}}}\right)^\tau & \forall v_j \in \mathcal{V}_i \\ \tau = 1; \frac{1}{2} \geq \bar{\theta}_{i,j} \geq 0 \\ \tau = 2; \frac{1}{2} \leq \bar{\theta}_{i,j} \leq 1 \end{cases} \quad (17)$$

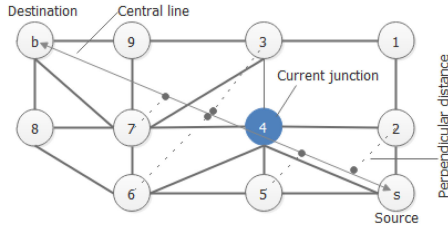


Fig. 3. Perpendicular distance.

the segment length probability distribution, denoted by $\tilde{\mathcal{L}}_i = (\tilde{\mathcal{L}}_{i,1}, \tilde{\mathcal{L}}_{i,2}, \dots, \tilde{\mathcal{L}}_{i,b_i})$, by the mass function as formulated in the following:

$$\bar{\mathcal{L}}_{i,j} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} / \sum_{\forall v_k \in \mathcal{V}_i} \sqrt{(x_k - x_i)^2 + (y_k - y_i)^2} \forall v_j \in \mathcal{V}_i \quad (21)$$

$$\tilde{\mathcal{L}}_{i,j} = \left(1 - \sqrt{e^{-\bar{\mathcal{L}}_{i,j}}}\right) / \sum_{\forall v_x \in \mathcal{V}_i} \left(1 - \sqrt{e^{-\bar{\mathcal{L}}_{i,x}}}\right) \forall v_j \in \mathcal{V}_i. \quad (22)$$

1) *Connectivity Distribution*: The connectivity of a given road segment $r_{i,j} \in \mathcal{E}_G$ is entirely hinged on the density of vehicles and the *interdistance* of vehicles. Hence, the density is computed based on multiple parameters, such as communication range δ , expected number of vehicles, the road segment length, and lane count. We defined the special density of vehicles by (23), where γ represents the lane count on the road segment, and $E_{i,j}^R$ denotes the expected number of vehicles on $r_{i,j}$, obtained by (11)

$$\hat{\rho}_{i,j} \triangleq \frac{\delta \cdot \gamma \cdot E_{i,j}^R}{|r_{i,j}|}. \quad (23)$$

To compute the segment connectivity, the road segment is divided into $2|r_{i,j}|/\delta$ equal blocks such that each block's length is $\delta/2$ meters. Thus, we obtained the probability of having k vehicle(s) on a block by the *Poisson* distribution, as expressed by (24). A block is connected if it holds at least one vehicle; this is obtained by (25). Thus, a road segment is connected, providing that its blocks are connected, as derived by (26)

$$\hat{R}_{i,j}(k) = \frac{(\hat{\rho}_{i,j})^k}{k!} e^{-\hat{\rho}_{i,j}} \quad (24)$$

$$\hat{R}_{i,j}(k > 0) = 1 - \hat{R}_{i,j}(0) = 1 - e^{-\hat{\rho}_{i,j}} \quad (25)$$

$$\tilde{\xi}_{i,j} = \prod_{c=1}^{2|r_{i,j}|/\delta} (1 - e^{-\hat{\rho}_{i,j}}) = (1 - e^{-\hat{\rho}_{i,j}})^{\frac{2|r_{i,j}|}{\delta}}. \quad (26)$$

Equation (26) expresses the relationships among the five parameters, which are: 1) the segment connectivity; 2) range of communication; 3) lanes count; 4) segment length; and 5) the expected vehicles count on a given road segment. To evaluate the performance of (26), we selected the following default settings. The number of vehicles is set to 5, the segment length is set to 1000m, transmission range is set to 100 m, and the

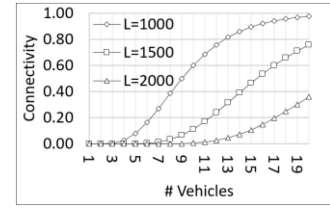


Fig. 4. Road segment connectivity varying road segment length and vehicles count.

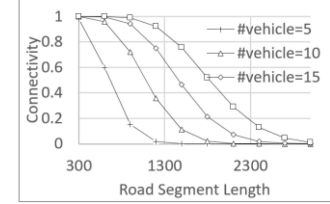


Fig. 5. Road segment connectivity varying the length of road segment and the vehicles count.

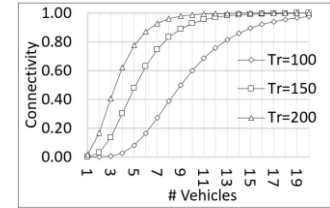


Fig. 6. Road segment connectivity. Varying transmission range and the number of vehicles.

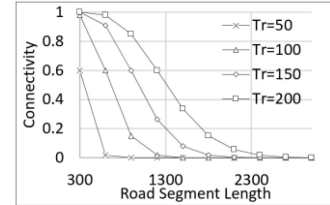


Fig. 7. Road segment connectivity. Varying transmission range and road segment length.

number of lanes for each roadway is set to 3. Three evaluation scenarios are considered. In the first scenario, both the segment length and the number of vehicles are varied. The number of vehicles varies from 1 to 20, and the length of the road segment is varied from 300 to 3000 m. The evaluation results are shown in Figs. 4 and 5. Fig. 4 shows that for a given segment length, the segment connectivity gets higher as the number of vehicles increases. While Fig. 5 shows that for a given number of vehicles, the connectivity gets lower as the segment length increases. In the second scenario, both the transmission range and the number of vehicles are varied. The evaluation results are shown in Fig. 6. It shows that for a given transmission range, the segment connectivity gets higher as the number of vehicles increases.

In the third scenario, both the transmission range and the road segment length are varied. The evaluation results are shown in Fig. 7. It shows that for a given transmission range, the connectivity gets higher as the segment length

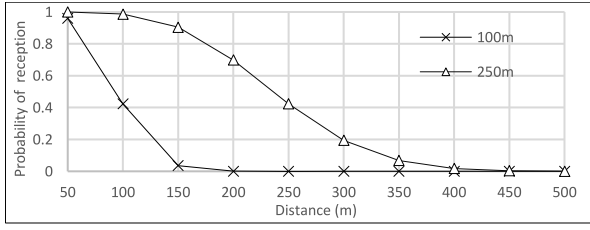


Fig. 8. Probability of receiving the packet over a given distance. Two values of transmission ranges are utilized 100 and 250 m.

decreases. Like [28], during simulations, we considered different transmission ranges 100, 250, and 1000 m for different devices.

C. IntraPath: Vehicle Selection

IntraPath selects the relay vehicles in a distributed manner. Thus, a vehicle n_i should know its neighbors, denoted by $\mathbb{N}_i$, via broadcasting beacon packet. To maintain its neighbor set, vehicle n_i periodically broadcasts beacon packets [5]. When it has a data packet to send, n_i selects a candidate set, denoted by $\mathbb{C}_i \subseteq \mathbb{N}_i$, which is computed according to a specific routing metric [29], [30]. From the candidate set $\mathbb{C}_i$, the sender vehicle selects one forwarder vehicle ($n_j \in \mathbb{C}_i$) to relay the packet. The process of selecting one forwarder from the candidate set is called the *candidate coordination*. This section presents a new routing metric that enhances the delivery ratio and reduces delivery delay in urban environments. We define our routing metric for each node n_i as a random variable $\tilde{Q}_i = \{\tilde{Q}_{i,1}, \tilde{Q}_{i,2}, \dots, \tilde{Q}_{i,q_i}\}$ by the mass function (27), which combines four probability distributions: 1) the speed difference $\tilde{S}_{i,j}$ (28); 2) the direction $\tilde{D}_{i,j}$ (29), shown at the bottom of the page; 3) the available buffer size $\tilde{M}_{i,j}$ (30); and 4) the signal fading $\tilde{F}_{i,j}$ (32). Considering the joint effect on (27), the moving direction has the greatest effect, while the other three distributions $\tilde{S}_{i,j} + \tilde{F}_{i,j} + \tilde{M}_{i,j}$ have an equal effect. Algorithm 2 outlines the selection of the next vehicle

$$\tilde{Q}_{i,j} = \frac{\tilde{D}_{i,j}(\tilde{S}_{i,j} + \tilde{F}_{i,j} + \tilde{M}_{i,j})}{\sum_{\forall n_x \in \mathbb{N}_i} \tilde{D}_{i,x}(\tilde{S}_{i,x} + \tilde{F}_{i,x} + \tilde{M}_{i,x})} \quad \forall n_j \in \mathbb{N}_i. \quad (27)$$

1) *Speed Difference Distribution*: The network performance can be enhanced by considering the least speed difference between two moving vehicles as both stay longer together and nearer to one another, which decreases the broadcasting of beacons between them. We will explain this later in the *neighbor assortment* (Algorithm 4). *Speed*

Algorithm 2 Vehicle Selection

Input: Sender vehicle n_i ; destination vehicle n_b .

Output: The next vehicle forwarder n_i with the highest priority. 1. VehicleSelection (n_i, n_b)

```

2. {
3.    $\forall n_i \in \mathbb{N}_i$  Do
4.     {
5.       Compute  $\tilde{S}_{i,j}$  by Eq. (28). // Speed Difference.
6.       Compute  $\tilde{D}_{i,j}$  by Eq. (29). // Moving Direction.
7.       Compute  $\tilde{M}_{i,j}$  by Eq. (30). // Buffer Size
         Distribution.
8.       Compute  $\tilde{F}_{i,j}$  by Eq. (32). // Signal Fading
         Distribution.
9.       Compute  $\tilde{D}_{i,j}(\tilde{S}_{i,j} + \tilde{F}_{i,j} + \tilde{M}_{i,j})$ .
10.    }
11.   $\forall n_i \in \mathbb{N}_i$  Do
12.    Compute  $\tilde{Q}_{i,j}$  by Eq. (27). // Vehicle priority.
13.  return vehicle  $n_i$  with the max  $\tilde{Q}_{i,j}$ .
14. }
```

difference distribution considers the similarity of speed between the moving vehicles such that greater priority values are allocated to those with the least difference of speed. The speed difference is modeled as a random variable such that the sender vehicle n_i has a speed difference random variable $\tilde{S}_i = \{\tilde{S}_{i,1}, \tilde{S}_{i,2}, \dots, \tilde{S}_{i,q_i}\}$ given by the mass function (28), where s_i denotes the current speed of vehicle n_i

$$\tilde{S}_{i,j} = e^{-\left(\frac{\sqrt{(s_i - s_j)^2}}{\max(s)}\right)} \bigg/ \sum_{\forall n_x \in \mathbb{N}_i} e^{-\left(\frac{\sqrt{(s_i - s_x)^2}}{\max(s)}\right)} \quad \forall n_j \in \mathbb{N}_i. \quad (28)$$

2) *Moving Direction*: We assume that the vehicle n_i located at $(\tilde{x}_i, \tilde{y}_i)$ is currently on the road segment r_{sd} that connects the start junction v_s and the end junction v_d . Vehicle n_i should select one relay vehicle $n_j \in \mathbb{N}_i$, which is currently in front of n_i . Vehicles in front of n_i are determined by angle β between two vectors as formulated in (29). In the case of the network partition, to take advantage of the vehicles that are not in the same direction as the current sender, HERO allows the sender vehicle to select the next forwarder from the opposite direction. This is implemented through the parameters Δ (same direction) and ∇ (opposite direction) as in (29). The smaller value of Δ allocates greater priority for the vehicles that are in the same direction as the sender. Likewise, a greater value of ∇ assigns a smaller priority for the vehicles that are in opposite directions.

3) *Buffer Size Distribution*: In order to reduce the queuing delay, that is, the time a packet waits in the queue until it

$$\beta = \cos^{-1} \left(\frac{(\tilde{x}_j - \tilde{x}_i)(x_d - \tilde{x}_i) + (\tilde{y}_j - \tilde{y}_i)(y_d - \tilde{y}_i)}{\sqrt{(x_d - \tilde{x}_i)^2 + (y_d - \tilde{y}_i)^2} \sqrt{(\tilde{x}_j - \tilde{x}_i)^2 + (\tilde{y}_j - \tilde{y}_i)^2}} \right) \bigg/ \pi$$

$$\tilde{D}_{i,j} = \begin{cases} \left(1 - \frac{\beta e^\beta}{1 + \beta e^\beta}\right)^\Delta; & 0 < \Delta \leq 1; \beta \leq \frac{1}{2} \\ \left(1 - \frac{\beta e^\beta}{1 + \beta e^\beta}\right)^\nabla; & 1 < \nabla \leq 5; \beta \geq \frac{1}{2} \end{cases} \quad \forall n_j \in \mathbb{N}_i \quad (29)$$

can be relayed, this distribution assigns higher priority for the vehicle with more available buffer (i.e., fewer packets in the queue). Let $\hat{m}$ be the buffer size and let $m_j \leq \hat{m}$ be the current number of packets in the buffer of vehicle n_j . The sender vehicle defines a random variable $\tilde{M}_i = \{\tilde{M}_{i,1}, \tilde{M}_{i,2}, \dots, \tilde{M}_{i,q_i}\}$ by the mass function

$$\tilde{M}_{i,j} = \frac{e^{-\frac{m_j}{\hat{m}}}}{1 + \frac{m_j}{\hat{m}}} = \frac{\hat{m} \cdot e^{-\frac{m_j}{\hat{m}}}}{\hat{m} + m_j} \quad \forall n_j \in \mathbb{N}_i. \quad (30)$$

4) *Signal Fading Distribution*: Signal fading is modeled as the *Nakagami distribution* with $m = 3$ as shown in (31) [25]. By applying $\Omega(d_{i,j})$ and $r_x(\delta)$ of (4) to (31), the probability of receiving the packet over the transmission distance $d_{i,j}$ is obtained by (32), where $d_{i,j}$ is the distance between sender vehicle n_i and receiver n_j (see Fig. 8). For a given sender vehicle n_i , signal fading to its neighboring vehicles is modeled as a random variable $\tilde{\mathcal{F}}_i = \{\tilde{\mathcal{F}}_{i,1}, \tilde{\mathcal{F}}_{i,2}, \dots, \tilde{\mathcal{F}}_{i,q_i}\}$. Note that signal fading is a function of the transmission distance and communication range. The transmission distance is obtained from the received signal strength indicator as in [31]

$$\begin{aligned} Pr(x > r_x) &= e^{-\frac{3}{\Omega} r_x} \sum_{i=0}^3 \frac{\left(\frac{r_x}{\Omega}\right)^{i-1}}{(i-1)!} \\ &= e^{-3\left(\frac{r_x}{\Omega}\right)} \left(1 + 3\left(\frac{r_x}{\Omega}\right) + 4.5\left(\frac{r_x}{\Omega}\right)^2\right) \end{aligned} \quad (31)$$

$$\tilde{\mathcal{F}}_{i,j} = e^{-3\left(\frac{d_{i,j}}{\delta}\right)^2} \left(1 + 3\left(\frac{d_{i,j}}{\delta}\right)^2 + 4.5\left(\frac{d_{i,j}}{\delta}\right)^4\right). \quad (32)$$

D. Neighbors Assortment and Data Routing

Data packets are routed from a source to a destination according to the procedures outlined in Algorithm 3. Given a sender vehicle n_k , which is currently traveling on the road segment $r_{ij} = (v_i, v_j)$. Let $\mathcal{H}(n_k, v_j)$ be the heading distance to junction v_j (i.e., the remaining distance until vehicle n_k reaches end junction v_j). We define $\hat{\mathcal{H}} = \delta/2$ as the heading distance threshold. If $\mathcal{H}(n_k, v_j)$ is smaller than threshold $\hat{\mathcal{H}}$, vehicle n_k starts selecting a new road segment r_{jx} by Algorithm 1. After selecting the road segment r_{jx} , the vehicle n_k searches for its neighbors in r_{jx} by sending the beacon packet and prioritizes the neighboring vehicles by Algorithm 2. On the other hand, as long as the heading distance is greater than the threshold, the vehicle picks its next hop from the same road segment.

To reduce the overhead and control the number of neighbors, HERO proposes the following algorithm, called *neighbor assortment* (Algorithm 4), which runs in two steps: 1) initialization and 2) prioritization. In the initialization step, when a sender vehicle n_i has a packet to send, it broadcasts a beacon packet to neighboring vehicles $\mathbb{N}_i$ and waits for one-hop vehicles (OHVs) to receive that beacon packet and send back a response. In the prioritization step, HERO utilizes the similarity function (33), which combines the speed difference (28) and the signal fading (32) to prioritize the OHVs according to their similarity score $\overleftrightarrow{\chi}_{i,j}$. The sender stores the ID of vehicles with a higher score $\overleftrightarrow{\chi}_{i,j} \geq 3/4$. The vehicles with

Algorithm 3 Data Routing

Input: The sender vehicle n_k has a packet in the road segment r_{ij} that joins two intersections v_i, v_j .
The destination vehicle n_b .

1. SendPacket(n_k)
2. {
3. if $\mathcal{H}(n_k, v_i)$ is smaller than $\hat{H}$ **Do**
4. {
5. Switch from r_{ij} to r_{jx} by **JunctionSelection**(v_i, v_b),
6. **ALGORITHM 1.**
7. Get $\mathbb{N}_k(r_{jx})$. // the neighbors of n_k in r_{jx} ; the segment $r_{j,x}$ joins the junctions v_j, v_x .
8. Select the next vehicle $n_v \in \mathbb{N}_k(r_{jx})$ by **VehicleSelection**(n_k, n_b),
9. **ALGORITHM 2.**
10. if n_v is found **Do**
11. **ReceivePacket**(n_v).
12. **else Do**
13. **EnqueuePacket**(n_k). n_k stores the packet to be retransmitted later.
14. **else** // $\mathcal{H}(n_k, v_j)$ is greater than $\hat{H}$.
15. {
16. Get the neighboring vehicles $\mathbb{N}_k(r_{ij})$ in the r_{ij} by sending beacons.
17. $n_v = \mathbf{VehicleSelection}(n_k, n_b)$ in r_{ij} by **ALGORITHM 2.**
18. **Goto** step line #8.
19. }
20. **end SendPacket.**
21. **ReceivedPacket**(n_v)
22. {
23. if n_v is the destination **then** the routing process is finished.
24. **else SendPacket**(n_v).
25. }
26. **end ReceivePacket.**

higher scores travel longer together and nearer to one another, which means that the sender will use them as fixed neighbors as long as possible. This strategy could reduce the overhead since the sender does not need to broadcast beacons within a period of time [namely, the keep trace-time (KTT)]. The overhead of HERO is analyzed in the next section

$$\begin{aligned} \overleftrightarrow{\chi}_{i,j} &= e^{-\left(\frac{\sqrt{(s_i - s_j)^2}}{\max(s)}\right)^2} - 3\left(\frac{d_{i,j}}{\delta}\right)^2 \\ &\quad \times \left(1 + 3\left(\frac{d_{i,j}}{\delta}\right)^2 + 4.5\left(\frac{d_{i,j}}{\delta}\right)^4\right). \end{aligned} \quad (33)$$

V. ANALYSIS

HERO adopts the flat-topology structure [12], [17], which indicates no extra overhead will incur to maintain the clusters (e.g., [32]) or zones (e.g., [14]), especially when vehicles are required to broadcast beacon packets periodically. The flat-topology structure is more efficient regarding overhead, especially for geographical areas with lower vehicle density if the topology changes rapidly. In a flat-topology structure, when the sender vehicle has a data packet, it broadcasts a beacon packet to check the surrounding vehicles. We assumed that each vehicle knows its neighbors or the so-called one-hop candidates (OHC) via broadcasting beacons. A sender vehicle picks a candidate vehicle (i.e., the forwarder) from its candidates to further forward the data packet. To reduce the communication overhead, HERO utilizes the *neighbor assortment* (Algorithm 4). In HERO, the overhead is computed based on beacon packets count that is required to be exchanged for selecting the next-hop forwarder, which is

Algorithm 4 Neighbors Assortment

Store the neighbor vehicles for a period of time. Assume that n_i receives a packet to be relayed.

```

1. Assortment( $n_i$ )
2. {
3.   if  $\mathbb{N}_i$  is empty Do
4.   {
5.     Send beacon packets.
6.     All OHVs send back a response.
7.      $n_i$  saves all OHVs.
8.     Run ALGORITHM 2 to select the next forwarder.
9.     //save the forwarders with a higher similarity score.
10.     $\forall n_i \in \mathbb{N}_i$  Do
11.    {
12.      Compute by  $\leftrightarrow \chi_{i,j}$  by Eq. (33).
13.      if ( $\leftrightarrow \chi_{i,j} \geq 3/4$ ) save  $n_j$  in  $\mathbb{N}_i$ .
14.    }
15.  }
16.  else
17.  {
18.    if keep trace time (KTT) is smaller than the
        threshold then
19.    Return one candidate from  $\mathbb{N}_k$  with the highest
        similarity score.
20.    else clear the  $\mathbb{N}_i$ ;
21.  }
22. }
```

correlated with the vehicles density and the range of communication. To compute the expected communication overhead, we computed the number of receivers of the beacon packets in each coordination stage (i.e., the selection of the next hope vehicle). This number is calculated by counting the number of OHCs when a sender vehicle transmits a beacon. The probability of having k OHCs within the communication range of the vehicle n_x is derived by (34), while the expected value of the communication overhead for one hop is expressed by (35)

$$\begin{aligned}
 \zeta(k) &= \frac{(\delta.\lambda)^k}{k!} e^{-\delta.\lambda} \\
 E[C_{\text{hop}}] &= \sum_{k=0}^{\infty} k.\zeta(k) = e^{-\delta.\lambda} \sum_{k=0}^{\infty} k. \frac{(\delta.\lambda)^k}{k!} \\
 &= \delta.\lambda . e^{-\delta.\lambda} \sum_{k=1}^{\infty} \frac{(\delta.\lambda)^{k-1}}{(k-1)!} \\
 &= \delta.\lambda . e^{-\delta.\lambda} \sum_{j=0}^{\infty} \frac{(\delta.\lambda)^j}{j!} \\
 &= \delta.\lambda . e^{-\delta.\lambda} . e^{\delta.\lambda} = \delta.\lambda.
 \end{aligned} \tag{34}$$

Furthermore, the communication overhead for a path $E[C_{\text{path}}]$ from a source vehicle to a destination vehicle is derived by (36), where $E[H]$ denotes the expected hops from a source to a destination, as derived in [33]

$$E[C_{\text{path}}] = E[C_{\text{hop}}].E[H]. \tag{36}$$

The probability mass function that characterizes k hops from a source to a destination is derived by (37), where $\mathcal{A}$ denotes the *border region*, derived by (38)

TABLE II
SIMULATION PARAMETERS

Parameter	Default value
Network size	4000m×4500m.
Junctions	12 (Traffic light 3s).
Road segments	17 (2000m horizontal, 1500m vertical).
Speed (km/h)	Max. 90 and Min. 0.
Packet size (bytes)	1024 for data 256 for control.
Com. range	500m.
Model	V2V.
Distance	2000m (from the source to the destination).
Storing time	5s (when the network is partitioned).
Retransmission attempts	7 times for RTS, MAC DCF 802.11[2].
# vehicles	400.
Channel data rate	2Mbps.
Radio Propagation	Nakagami-m.
Δ, ∇	$\Delta = 0.1; \nabla = 3; \text{ Eq.(29).}$
w_ϕ, w_ξ	$w_\phi = 0.6$ and $w_\xi = 0.4; \text{ Eq.(14).}$

$$\eta(k) = \left(e^{-\lambda\pi\delta^2(k-1)^2} - e^{-\lambda\pi\delta^2k^2} \right) \left(1 - e^{\lambda\mathcal{A}/2} \right)^{k-1} \tag{37}$$

$$\mathcal{A} = \delta^2. \left(\frac{\pi - 2}{4} \right) + \mathcal{L}. \left(\theta - \frac{\sin(2\theta)}{2} \right). \tag{38}$$

The expected hops are obtained by

$$\begin{aligned}
 E[H] &= \sum_{k=0}^{\infty} k.\eta(k) \\
 &= k \sum_{k=0}^{\infty} \left(\left(e^{-\lambda\pi\delta^2(k-1)^2} - e^{-\lambda\pi\delta^2k^2} \right) \left(1 - e^{\lambda\mathcal{A}/2} \right)^{k-1} \right).
 \end{aligned} \tag{39}$$

VI. SIMULATION AND EVALUATION RESULTS

We have conducted extensive simulation experiments to assess the performance of our proposed protocol.

A. Environment

We used VSIM [23], which is a 2-D simulator provides an acceptable animation and visualization of experimental runs and information about vehicles and road network, available in the link.³

1) *Settings*: The default simulation parameters are listed in Table II. We used a road network of size 4000 m × 4500 m that contains 12 intersections (with traffic light of 3 s) and 17 segments, each segment with two roadways, each of two lanes. The starting position of the vehicles is randomly distributed over the road segments. The max allowed speed of the road segment is 90 km/h, while a min speed is 0 km/h. Vehicles gradually decrease the speed when approaching the heading intersection and line-up in a queue to take their turn. The *InterPath* is controlled through two turning parameters, w_ϕ to control the shortest-distance distribution, while w_ξ is to control the CD. Greater value of the turning parameter assigns higher priority for the corresponding distribution. We set the values of tuning parameters w_ϕ and w_ξ to 0.6 and 0.4, respectively. The impact of these parameters is enlightened latter in the following section. On the other hand, *IntraPath* is enhanced by two tuning parameters: 1) Δ (same direction) and 2) ∇ (opposite

³<https://github.com/howbani/VSIM>

direction) to take the advantage of the vehicles which are not in the same direction.

2) Metrics:

1) *Communication Overhead*: The number of control packets includes beacons, ACK, and the control packets that are intended to maintain the dissemination structure such as cluster or zone.

2) *Delivery Latency*: The delay time from end to end in seconds. It is calculated from the time in which a data packet is generated at the source until it is received by the destination vehicle, including the waiting delay, propagation latency or transmission delay, and queuing latency.

3) *Packet Success Ratio*: The ratio of successfully delivered packets (i.e., to their destinations) to the packets that are generated at the source vehicles.

3) *Comparison Approaches*: To meticulously evaluate HERO, we selected two different protocols for comparison:

1) clustering structure and 2) flat structure. The first protocol is a clustered-based protocol called MOZO [14], while the second protocol is a flat structure called BRAVE [16]. BRAVE and MOZO are explained in Section II. MOZO is a new clustering-based protocol and outperformed the existing protocols such as CBDRP [32], while BRAVE is selected to represent the nonclustering approaches because BRAVE outperformed SAR, ASTAR, GPCR, and GeOpps, mentioned in [14] and [16]. Due to rapid topology changes, MOZO incurs more communication overhead for building, maintaining, merging, and splitting the zones. Conversely, during data routing, MOZO has lower overhead since member vehicles exclusively communicate with the captain vehicle. One drawback of MOZO is that the road segment is selected by considering the shortest distance without paying attention to the connectivity of the segment, which in some cases makes it possible to drop a packet or slow down its delivery. Different from MOZO, the flat structure BRAVE adopts an opportunistic mechanism to diminish the communication overhead that requires to maintain a hierarchical data dissemination structure.

B. Experimental Results

The results reported are the average of ten independent runs for the same configuration by varying the following parameters, the distance from the source vehicle to the destination vehicle, vehicle's number in the road network, the count of packets, and the speed of the vehicles.

1) *Varying Data Packet Delivery Distance*: The default number of vehicles is set to 400, while the source vehicle number is set to 100, which are randomly selected at a distance between 600 and 3000 m from the destination vehicles, as in [14]. In each simulation run, 100 data packets are generated.

Delivery Latency: Fig. 9 presented the simulation results of delivery latency varying distance values. We concluded that the longer distance between the source vehicle and the destination vehicle undoubtedly leads to more intermediate hops, which increases the latency. The reported results showed that the protocols employed the flat structure (i.e., HERO and BRAVE)

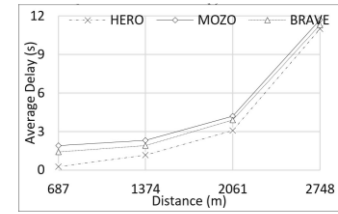


Fig. 9. Average delivery latency varying distance.

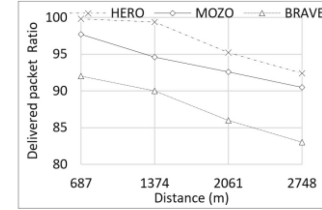


Fig. 10. Packet success ratio varying distance.

outperform the protocols that employed the clustered structure (i.e., MOZO). This is because the flat-structure protocols do not require construction or maintenance. HERO showed faster time than BRAVE and MOZO, owing to the two reasons as follows. Both BRAVE and MOZO employ the shortest path (*Dijkstra's algorithm*) to pick up the next-hop segment without taking into account the segment connectivity. Such selection is not always effective because the connectivity of the shortest segment cannot be always guaranteed. In contrast, HERO aggregates both of the segment connectivity and the shortest distance in the heuristic function (14), which allows HERO to select the road segment professionally. Besides, BRAVE opportunistically selects the next vehicle considering the response time of the ACK without taking into account the speed of the vehicle and the fading of the signal. BRAVE delivers the data packets faster than MOZO, since MOZO's structure needs maintenance and also the complexity of the forwarding mechanism in MOZO, which requires relaying the packet of cluster's members to cluster's captain first, and then from the cluster's captain to the next-hop member.

Packet Success Ratio: Fig. 10 presents the simulation results of the packet success ratio with varying different distance-values. We concluded that the long distance between the source and destination vehicles gradually decreases the packet success ratio. This primarily due to the reason that the network may subject to partitions with longer distances, which leads to drop the packet. Network partition probability gets higher when the value of the distance is increased due to numerous reasons, including but not limited to channel fading and distribution of vehicles. Thus, HERO outperforms BRAVE and MOZO due to the connectivity function that imposes the data packets to be routed through road segments with satisfactory density and shorter distance. Likewise, different from BRAVE and MOZO, our proposed protocol, HERO, rationally utilizes the network's resources through exploiting the vehicles in both directions of the road way, in case of two or more lanes, which surges the packet success ratio, particularly in the network with lower density. This is implemented by

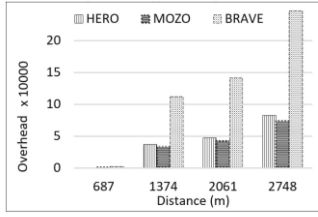


Fig. 11. Communication overhead varying distance.

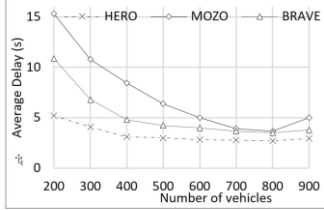


Fig. 12. Average delivery latency varying number of vehicles.

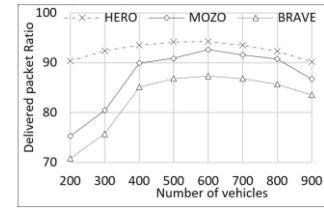


Fig. 13. Packet success ratio varying vehicles count.

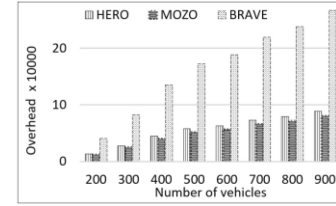


Fig. 14. Communication overhead varying vehicles count.

the two parameters: 1) Δ (same direction) and 2) ∇ (opposite direction) in (29). MOZO demonstrates a better success ratio compare to BRAVE. This is because the cluster's header in MOZO interacts with more vehicles, which, in turn, gives MOZO a higher chance to retransmit the data packet before depleting the predefined retransmission attempts.

Communication Overhead: Fig. 11 presents the simulation results of communication overhead varying different values of distance. It shows that the communication overhead gets greater as the distance from the source vehicle to the destination vehicle increases. This happens due to the necessity of picking more candidate vehicles, and hence more control packets are generated all the way until the data packet arrives its destination. HERO and MOZO incurred similar communication overhead, while BRAVE incurred more overhead since BRAVE first transmits the data packet to the candidate vehicles and then picks up a candidate to further disseminates the data packet toward the destination. HERO shows good performance, nearly equal to the clustered-based protocols due to utilizing the *neighbor assortment* (33), which saves the candidate's information based on the score of similarity, which highly reduces the periods of discovering nearby candidate vehicles. MOZO exhibits a smaller communication overhead since the clustered-structure reduces many of control packets during the data dissemination since the members openly contact with the cluster's header, and no need for control packets to discover the next-hop vehicle (candidate vehicle). However, MOZO suffered a lot of overhead for constructing, maintaining, merging, and splitting the moving zones.

2) Varying Vehicles Number: In this evaluation scenario, vehicles count varies from 200 to 900, from which 100 source vehicles are selected, each source vehicle picks up a destination vehicle within a distance of 2000 m, and each source vehicle generates one data packet.

Delivery Latency: Fig. 12 shows the simulation results of latency varying the vehicles count. It shows that a large number of vehicles highly increase the connectivity of the network. This certainly reduces the latency since the *carry-and-forward*

times are reduced. With lower density, the vehicles carry the packet multiple times, and also it takes longer for the data packet to be disseminated. Our protocol, HERO, reduces the times of *carry-and-forward*, as it continuously elites the segment that has greater connectivity and a shorter distance, which significantly reduces the chance of a packet to travel via a partitioned road segment. For this reason, HERO attains faster delivery of packets than BRAVE and MOZO. On the contrary, BRAVE and MOZO used Dijkstra's algorithm to pick up the road segment with the shortest distance without taking into account the segment connectivity. This seriously consumes the number of carry-and-forwards and hence makes the delivery of packets slower.

Packet Success Ratio: Fig. 13 shows the simulation results of the packet success ratio varying the vehicles count. It indicates that the delivery rate increases with the count of vehicles then decreases, following a bell curve. As the number of vehicles increases from 200 to 600, the connectivity of the network increases, so the chance of encountering a network partition decreases. This undoubtedly increases the delivery success rate. Also, the results indicated that network connectivity gets higher when the number of vehicles between 700 and 900. But this likely causes channel conflict and may lead to more collisions, so more packets will be dropped. HERO demonstrated a better packet success ratio compared to BRAVE and MOZO. This is partially due to the connectivity function (26) and the segment length mass function (22). These two mass functions balanced the traffic jam and network connectivity. In longer road segments, especially when the number of vehicles is large, vehicles may be fairly distributed rather than crowded. In such a case, the packet is better to travel through a longer segment than to travel in a short segment. In HERO, the SDD $\Phi_{i,j}$ is obtained by combining two mass functions, as in (15), the direction angle $\theta_{i,j}$ and perpendicular distance $\psi_{i,j}$. The direction distribution assigns higher priority for junctions toward the destination, while the perpendicular distance distribution assigns higher priority for junctions that are closer to the centerline between the source and the destination.

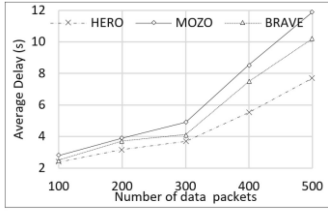


Fig. 15. Average delivery latency varying packets count.

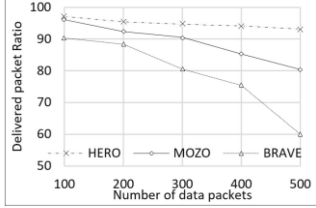


Fig. 16. Packet success ratio varying packets count.

Communication Overhead: Simulation results of overhead varying vehicles count are presented in Fig. 14. Obviously, more vehicles in a road segment increase the communication overhead since the candidates in each hop are increased. For the clustered-based protocol MOZO, more control packets are generated to maintain the MOZOs, which definitely raises the network overhead. Conversely, the overhead in flat-based protocols HERO and BRAVE is increased since more beacon packets are generated (i.e., more candidate vehicles). HERO and MOZO incurred similar overhead, which is lower than the overhead incurred by BRAVE. The detailed reasons for this have been listed so far in the previous section.

3) *Varying Packet Count:* We evaluated HERO's performance by varying packet count, namely, between 100 and 500 packets. Vehicles count is set by default to 400, while the separation distance from the source vehicle to the destination vehicle is set by default to 2000 m. The packets are instantaneously generated at the same moment.

Delivery Latency: Fig. 15 shows the simulation results of delivery latency varying the number of packets. BRAVE and HERO showed lower latency than MOZO partially due to the reasons clarified in the previous section. Furthermore, to reduce the queuing delay (i.e., the time a packet waits in the queue until it can be relayed), HERO utilizes the mass function of buffer size [i.e., (30)], which assigns higher priority for the vehicles with more available buffers. The next-hop vehicle in HERO is selected based on the heuristic function (27), which combines four probability distributions: 1) the speed difference $\tilde{S}_{i,j}$ (28); 2) the direction $\tilde{D}_{i,j}$ (29); 3) the available buffer size $\tilde{M}_{i,j}$ (30); and 4) the signal fading $\tilde{F}_{i,j}$ (32). Such quantities are not considered in BRAVE and MOZO.

Packet Success Ratio: Simulation results for packet success ratio varying the packets count in the network are presented in Fig. 16. More packets in the network certainly consumes the wireless medium and increases the contention. This definitely reduces the success ratio of packets. HERO demonstrated better performance compared to BRAVE and MOZO partially due to the reasons clarified in the previous section.

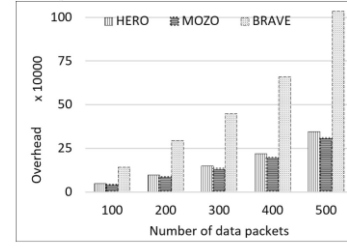


Fig. 17. Communication overhead varying packets count.

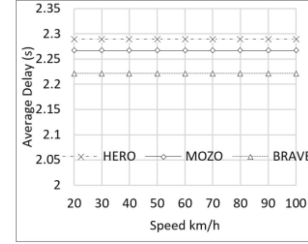


Fig. 18. Average delivery latency varying vehicle's speed.

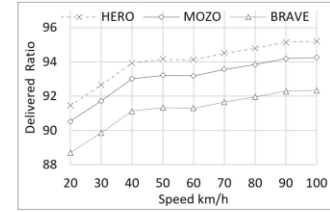


Fig. 19. Packet success ratio varying vehicle's speed.

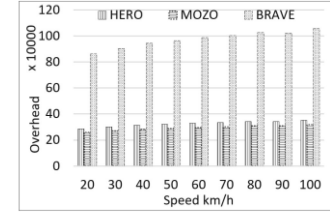


Fig. 20. Communication overhead varying vehicle's speed.

Communication Overhead. Fig. 17 shows the simulation results of overhead varying the packets count. It shows that more packets generated in the network definitely rise the overhead partially due to the reasons clarified in the previous section.

4) *Varying Vehicle Speed:* In this scenario, the number of vehicles is set to 400 by default. The average speed of vehicles varies between 20 and 100 km/h. For each average speed, 1000 data packets are generated randomly within 2000 m of the distance. The other parameters are defined as shown in Table II. As a result, vehicle speed has no significant impact on the delivery delay as shown in Fig. 18. However, the average speed has a slight impact on the delivery rate and overhead of the network, as shown in Figs. 19 and 20, respectively. For a given density, especially when the network is partitioned, the rapid network changes may lead to fast network recovery.

This can cause an increase in both the delivery ratio and the network overhead.

VII. CONCLUSION

We presented a distributed routing protocol for the vehicular urban environment, HERO, which comprises two heuristic functions: 1) *Interpath* and 2) *Intrapath*, to optimize the selection of road segments and vehicles on road segments, respectively. The *Interpath* was designed to select a routing path that consists of multiple road segments with a shorter distance and higher connectivity. This was implemented through two probability distributions, namely, the SDD and the CD. Each distribution considers multiple quantities that capture the physical property, which enhances the routing decision. SDD considers the direction angle, the perpendicular distance, and segment length while CD considers the communication range, lane count on the road segment, the length of the segment, and the expected vehicles count. On the other hand, the heuristic function of the *Intrapath* prioritizes vehicles in the road segment based on four routing quantities: 1) speed difference; 2) the vehicle moving direction; 3) buffer size; and 4) signal fading.

Considering the mechanism of multicriteria routing can highly enhance the performance of VANET, however, two main difficulties are faced during working in this article. First, how to mathematically model each criterion, while the second difficulty is how to aggregate these conflict criteria in order to obtain an optimal routing decision. Despite HERO supports V2V communication, it can be modified to support V2I communication by allowing the vehicles to hand the packets to RSUs at junctions rather than searching for a forwarder vehicle on other neighbor road segment.

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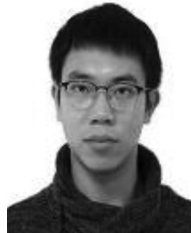
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